

EE121 Discussion 6

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Agenda

- X-ary Pulse Amplitude Modulation (PAMX)
 - Probability of Error
 - Soft-Decision
- Frequency Shift Keying (FSK) when things aren't perfect

Probability of Error PAM2

$$y(t) = x(t) + n(t), n(t) \sim \mathcal{N}(0, N_0)$$

Signal Model for PAMX

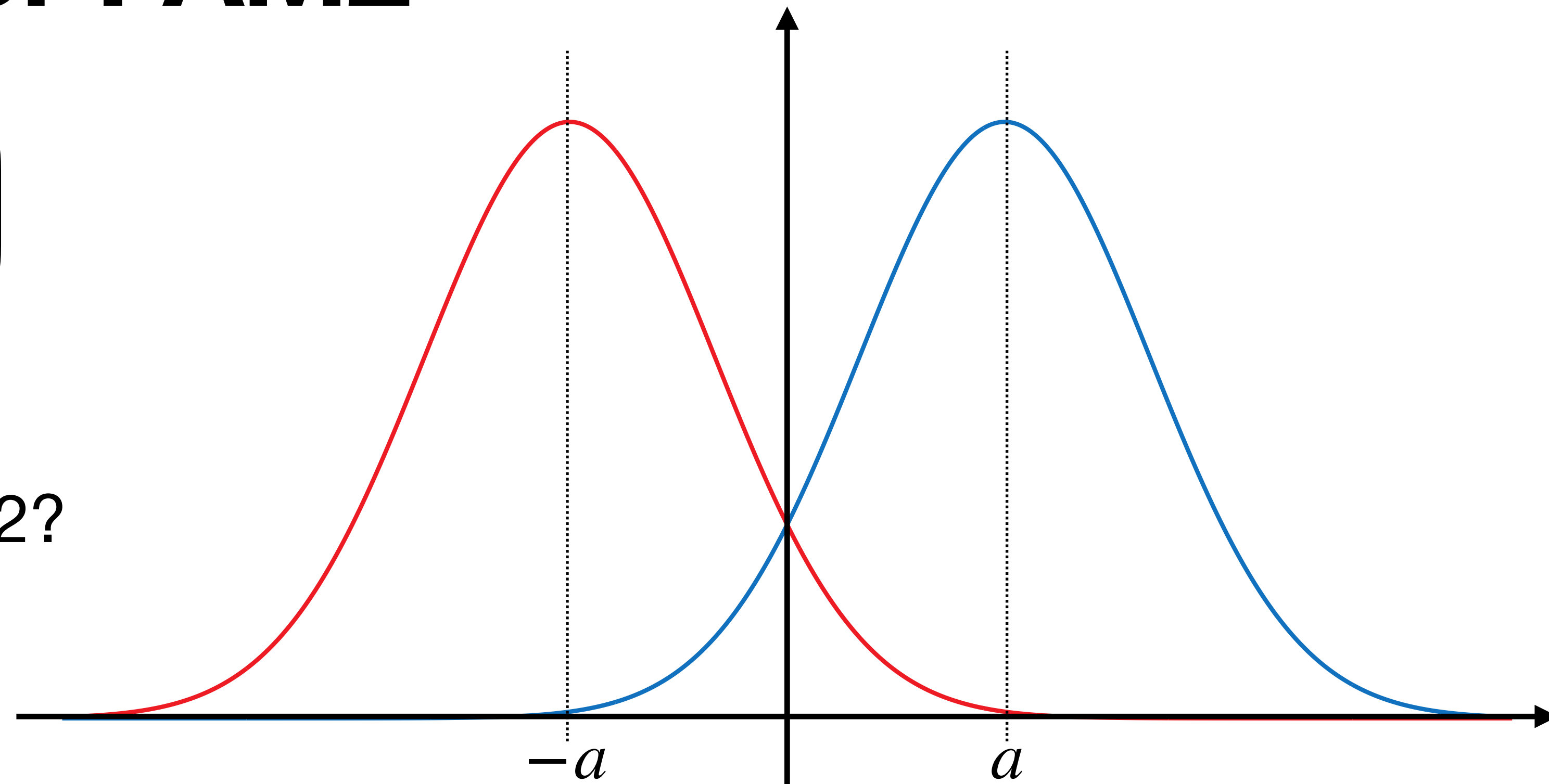
What are my hypotheses for PAM2?

H_0 : “0” was sent at nT_b

H_1 : “1” was sent at nT_b

Decision regions for MLE?

$$\hat{a} = 0$$

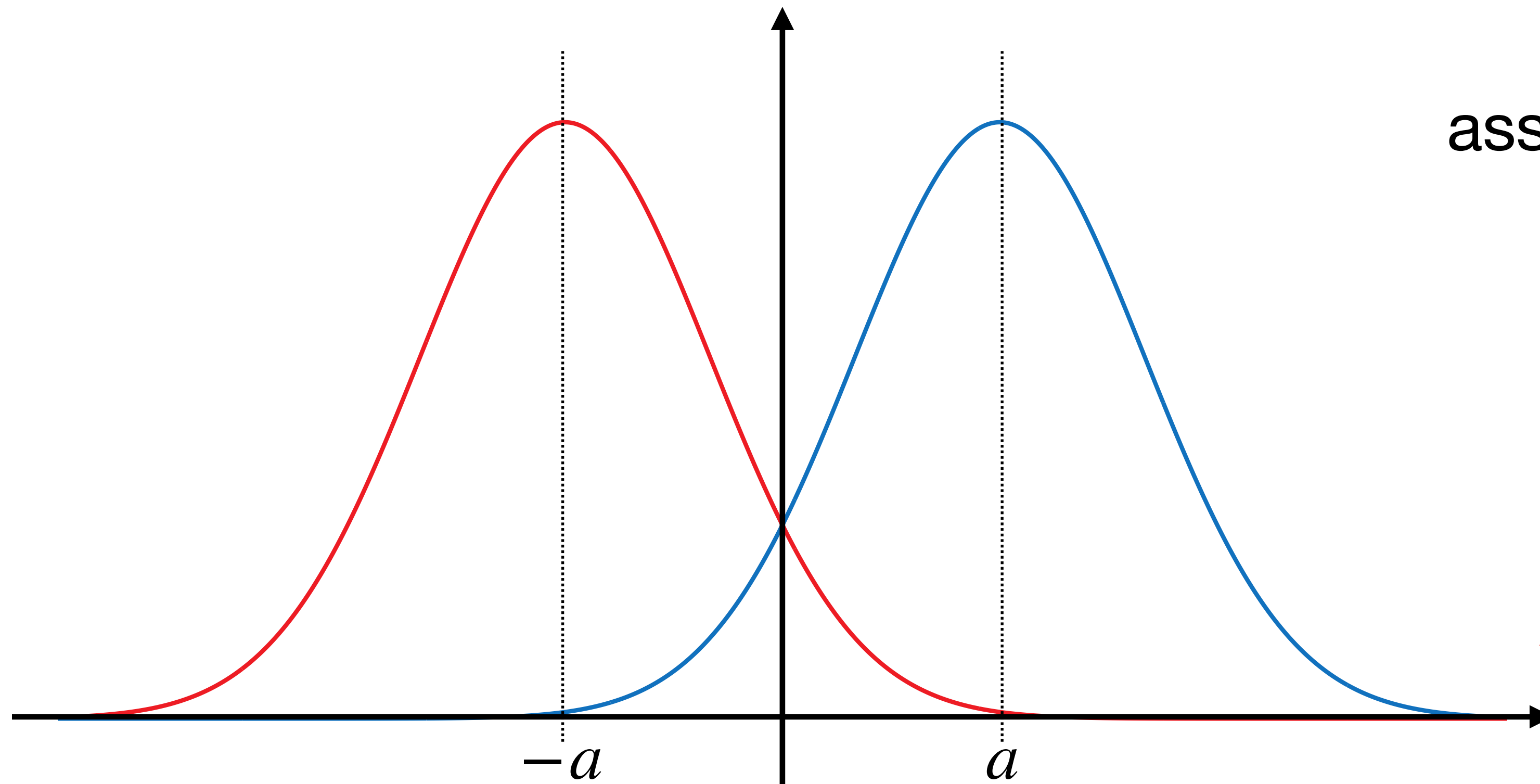


PAM2

$$y(nT_b) = a \rightarrow 1$$

$$y(nT_b) = -a \rightarrow 0$$

Probability of Error PAM2



PAM2

$$y(nT_b) = a \rightarrow 1$$

$$y(nT_b) = -a \rightarrow 0$$

What's the probability of error assuming 1 and 0 are equally likely?

H_0 : "0" was sent at nT_b

H_1 : "1" was sent at nT_b

$$P[H_0 | 1] = P(\hat{a} > 0) = Q\left(\frac{a}{\sqrt{N_0}}\right)$$

$$P[H_1 | 0] = P(\hat{a} < 0) = Q\left(\frac{a}{\sqrt{N_0}}\right)$$

$$P_e = \frac{1}{2}P[H_0 | 1] + \frac{1}{2}P[H_1 | 0]$$

Probability of Error PAM4

$$y(t) = x(t) + n(t), n(t) \sim \mathcal{N}(0, N_0)$$

Signal Model for PAMX

What are my hypotheses for PAM4?

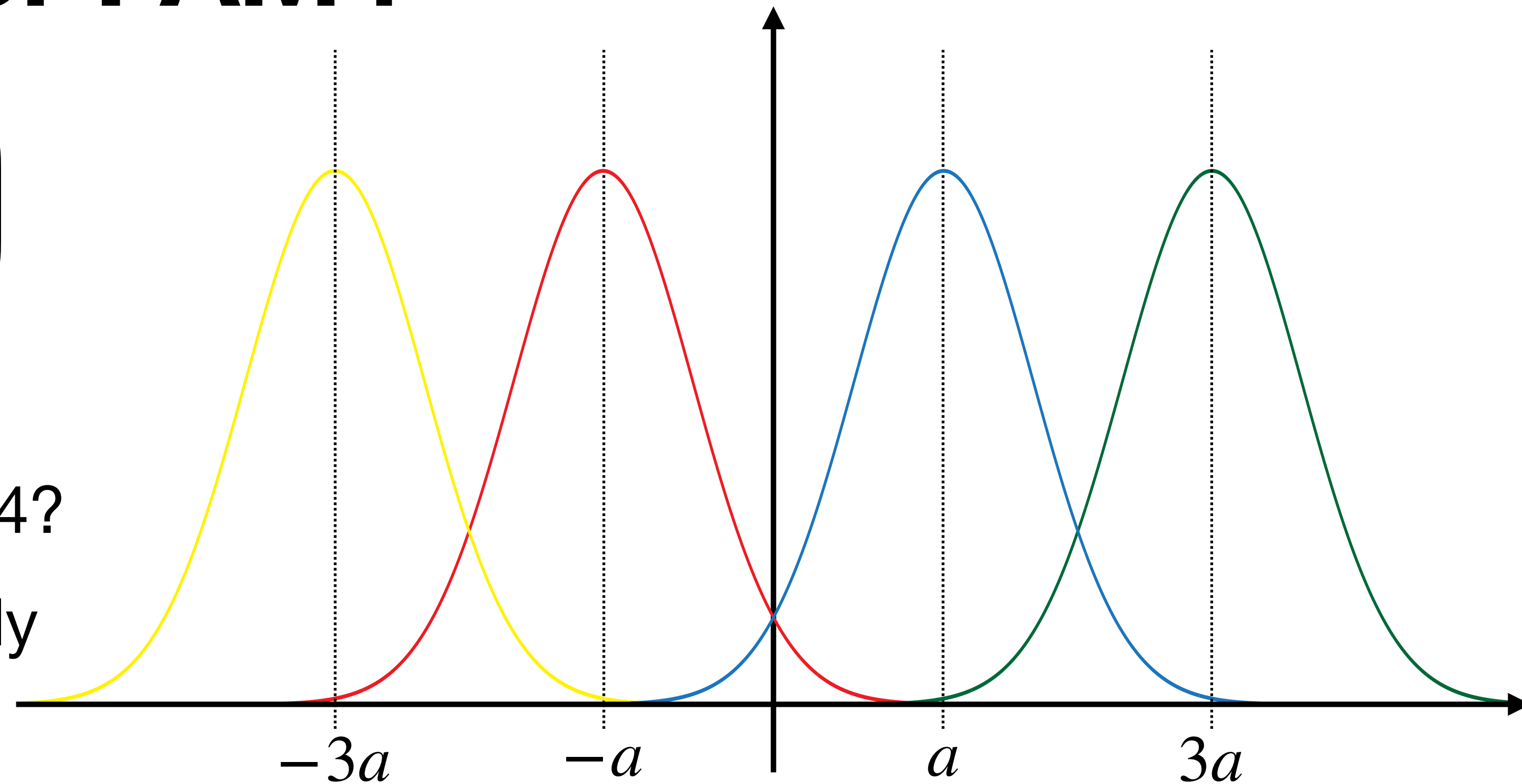
Decision Regions for equally likely symbols?

H_0 : $-3a$ was sent at nT_b

H_1 : $-a$ was sent at nT_b

H_2 : a was sent at nT_b

H_3 : $3a$ was sent at nT_b

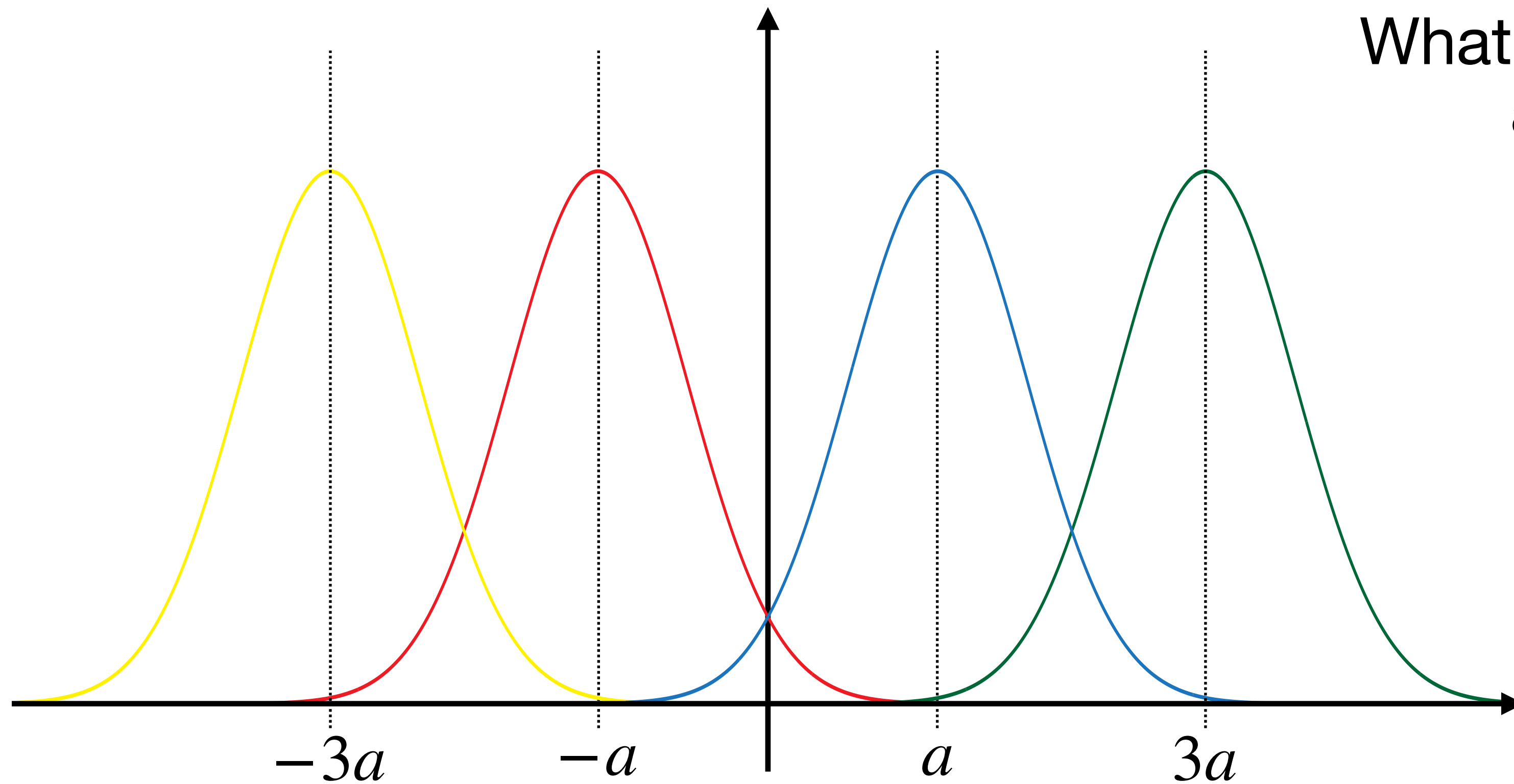


PAM4

$y(nT_b) = 3a \rightarrow 11$ $y(nT_b) = -a \rightarrow 00$

$y(nT_b) = a \rightarrow 10$ $y(nT_b) = -3a \rightarrow 01$

Probability of Error PAM4



PAM4

What's the probability of error assuming all symbols are equally likely?

H_0 : $-3a$ was sent at nT_b

H_1 : $-a$ was sent at nT_b

H_2 : a was sent at nT_b

H_3 : $3a$ was sent at nT_b

$y(nT_b) = 3a \rightarrow 11$ $y(nT_b) = -a \rightarrow 00$

$y(nT_b) = a \rightarrow 10$ $y(nT_b) = -3a \rightarrow 01$

Probability of Error PAM4

What's the probability of error assuming all symbols are equally likely?

H_0 : $-3a$ was sent at nT_b

H_1 : $-a$ was sent at nT_b

H_2 : a was sent at nT_b

H_3 : $3a$ was sent at nT_b

$$P_e = \sum \frac{1}{4} P_e[H_i] = \frac{6}{4} Q\left(\frac{a}{\sqrt{N_0}}\right)$$

$$P[y > -2a \mid x = -3a] = Q\left(\frac{a}{\sqrt{N_0}}\right)$$

$$P[y > 0 \text{ or } < -2a \mid x = -a] = 2Q\left(\frac{a}{\sqrt{N_0}}\right)$$

$$P[y > 2a \text{ or } < 0 \mid x = a] = 2Q\left(\frac{a}{\sqrt{N_0}}\right)$$

$$P[y > -2a \mid x = 3a] = Q\left(\frac{a}{\sqrt{N_0}}\right)$$

Do you see the pattern?

$$Q\left(\frac{a}{\sqrt{N_0}}\right)$$

PAM2

HW9

PAM3

$$\frac{6}{4}Q\left(\frac{a}{\sqrt{N_0}}\right)$$

PAM4

$$f(X)Q\left(\frac{a}{\sqrt{N_0}}\right)$$

PAMX

Soft Decisions with PAM4

The received signal has a level $\hat{a} = 1.25a$. Use a soft decision to estimate the symbol. ($a = 1, \sigma = 1$)

$$P[-3a] = 0.3$$

$$P[-a] = 0.2$$

$$P[a] = 0.1$$

$$P[3a] = 0.4$$

$$H_0: -3a \text{ was sent at } nT_b$$

$$H_1: -a \text{ was sent at } nT_b$$

$$H_2: a \text{ was sent at } nT_b$$

$$H_3: 3a \text{ was sent at } nT_b$$

$$P[H_i | \hat{a}] = \frac{p(\hat{a} | i)P[i]}{p(\hat{a})} = \frac{p(\hat{a} | i)P[i]}{\sum_{j=0}^{M-1} p(\hat{a} | j)p(j)}$$

Symbol Probability

Hypotheses

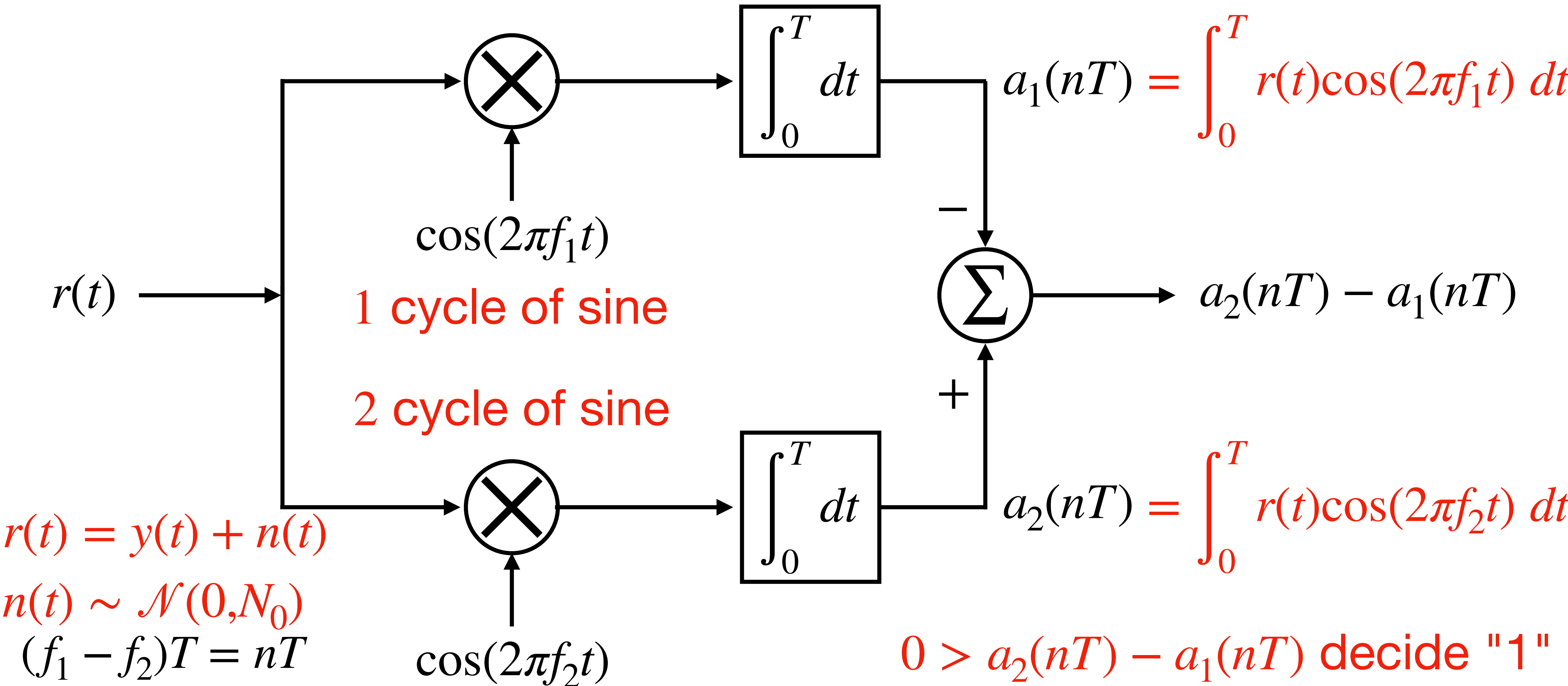
Soft Decision Rule

$$P[H_0 | \hat{a} = 1.25] = \frac{p(1.25 | -3)P[-3]}{p(1.25 | 3)P[3] + p(1.25 | 1)P[1] + p(1.25 | -1)P[-1] + p(1.25 | -3)P[-3]}$$

$$P[H_0 | \hat{a} = 1.25] = 0, P[H_1 | \hat{a} = 1.25] = 0.006, P[H_2 | \hat{a} = 1.25] = 0.094, P[H_3 | \hat{a} = 1.25] = 0.9$$

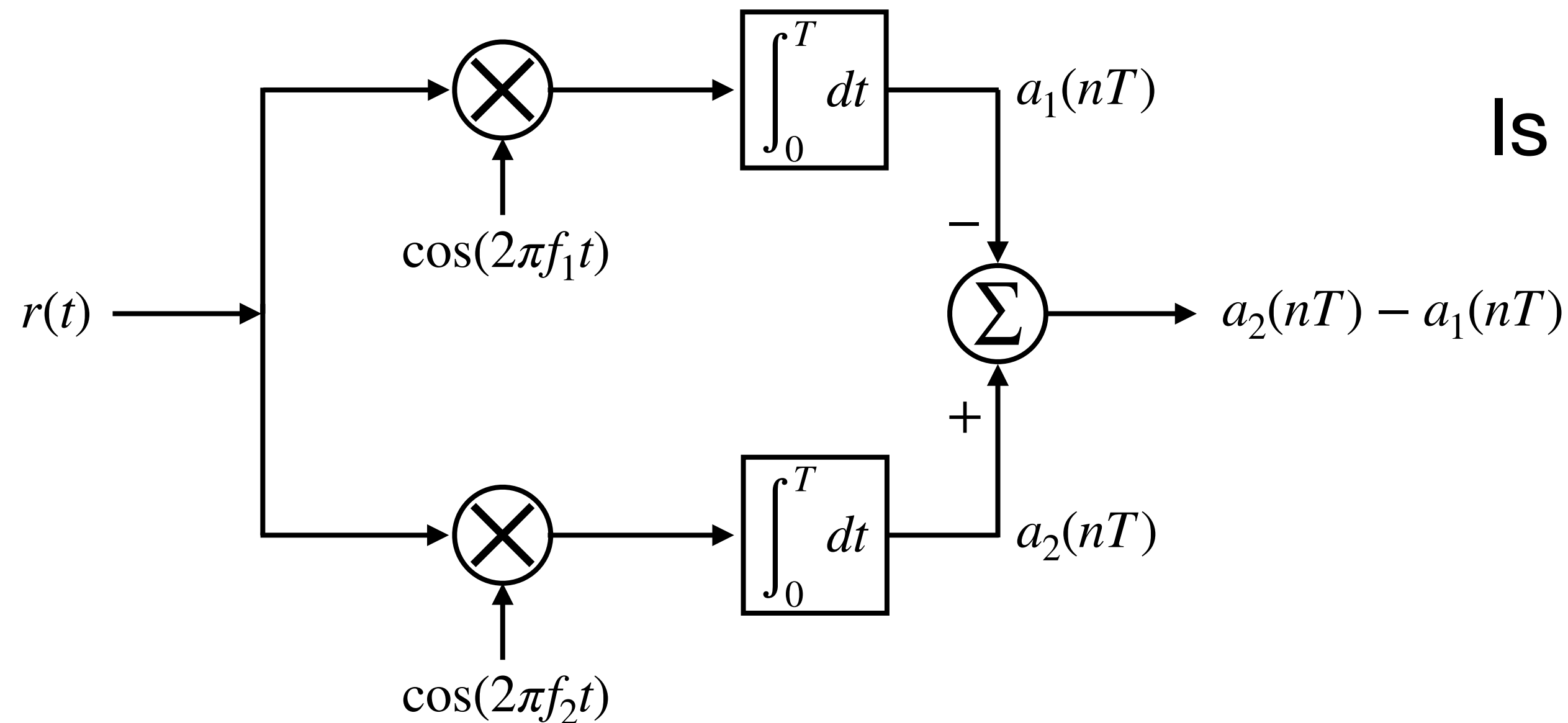
FSK2 (Coherent Detection)

$$\int_0^T \cos(2\pi f_1 t) \cos(2\pi f_2 t) dt = 0$$



What if $(f_1 - f_2)T$ is not an integer?

What are a_1 and a_2 ?



Is the noise correlated between a_1 and a_2 ?

Let's study a_1 and a_2 .

$$r(t) = A \cos(2\pi f_2 t) + n(t), \quad n(t) \sim \mathcal{N}(0, N_0)$$

$$a_2(nT) = \int_0^T r(t) \cos(2\pi f_2 t) dt = \frac{AT}{2} + \int_0^T n(t) \cos(2\pi f_2 t) dt$$

$$a_1(nT) = \int_0^T r(t) \cos(2\pi f_1 t) dt = \boxed{\int_0^T A \cos(2\pi f_2 t) \cos(2\pi f_1 t) dt} + \int_0^T n(t) \cos(2\pi f_1 t) dt$$

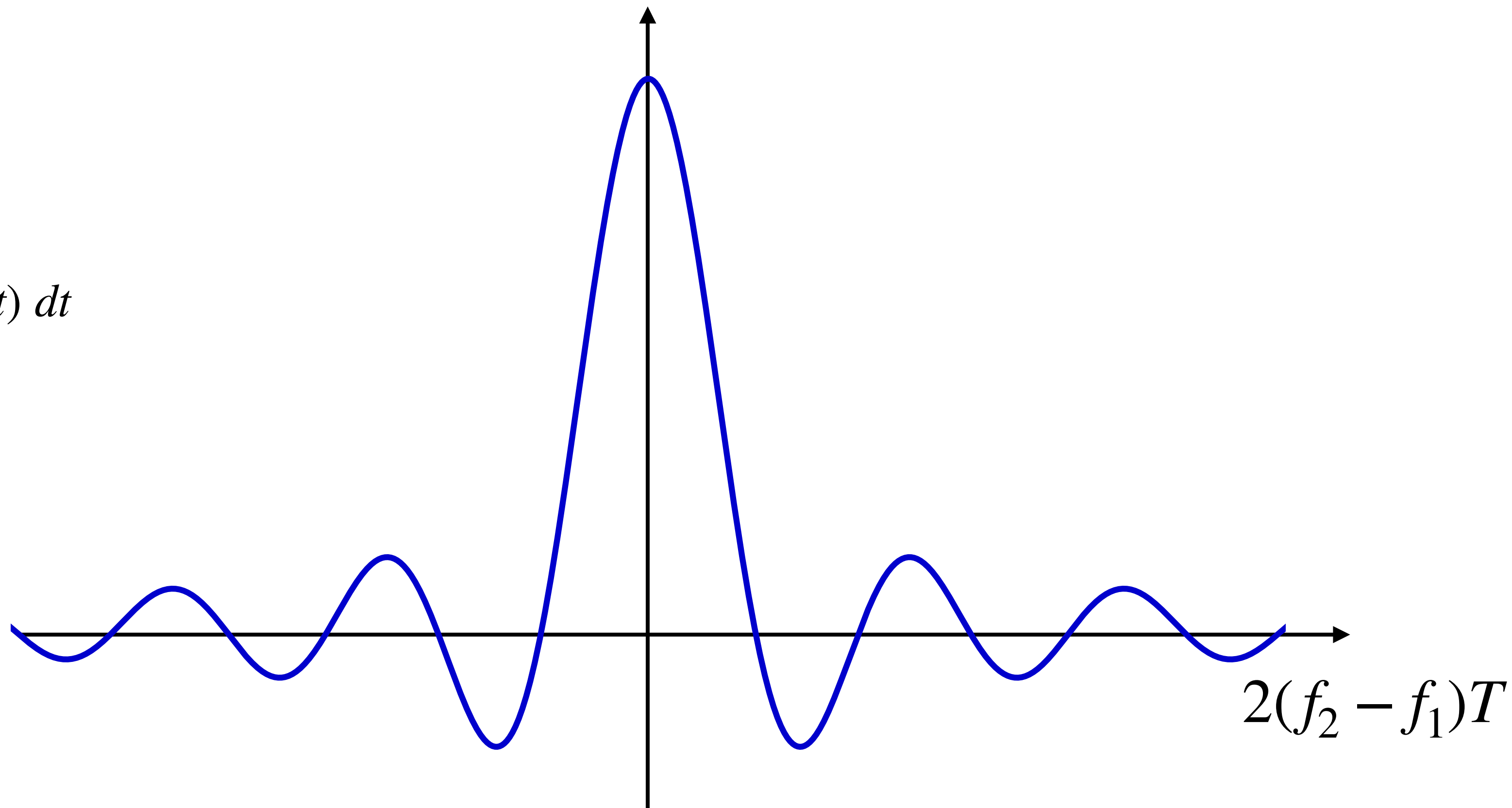
$$\approx \frac{A}{2} \int_0^T \cos(2\pi(f_2 - f_1)t) dt = \frac{A}{2} \frac{\sin(2\pi(f_2 - f_1)t)}{2\pi(f_2 - f_1)} \Bigg|_0^T = \frac{A}{2} \frac{\sin(2\pi(f_2 - f_1)T)}{2\pi(f_2 - f_1)} = \frac{AT}{2} \text{sinc}(2(f_2 - f_1)T)$$

What if $(f_1 - f_2)T$ is not an integer?

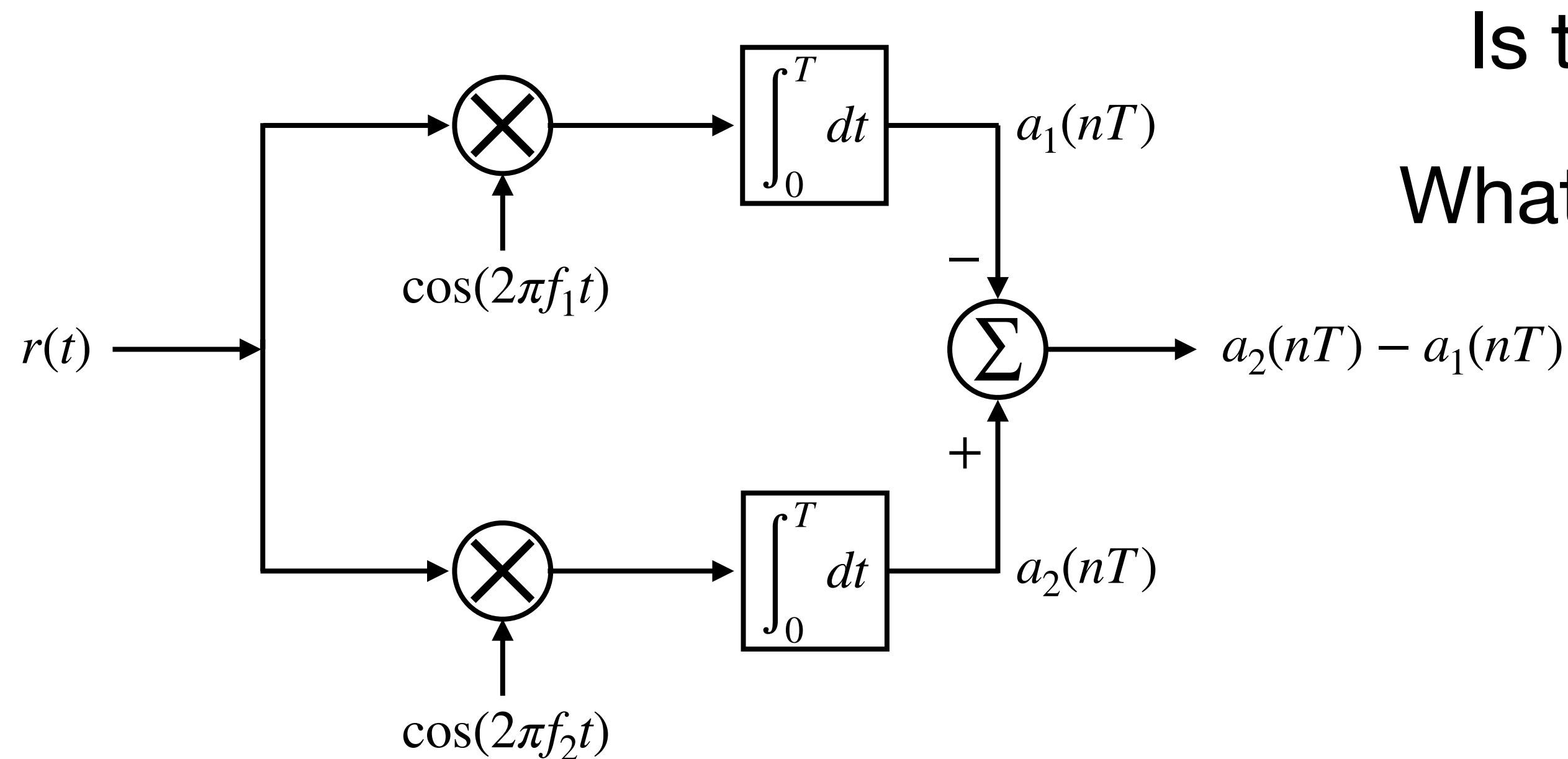
$$r(t) = A \cos(2\pi f_2 t) + n(t), \quad n(t) \sim \mathcal{N}(0, N_0)$$

$$a_2(nT) = \frac{AT}{2} + \int_0^T n(t) \cos(2\pi f_2 t) dt$$

$$a_1(nT) \approx \frac{AT}{2} \text{sinc}(2(f_2 - f_1)T) + \int_0^T n(t) \cos(2\pi f_1 t) dt$$



What if $(f_1 - f_2)T$ is not an integer?



Is the noise correlated between a_1 and a_2 ?

What's the variance of the noise at the output?

$$N_1 = \int_0^T n(t) \cos(2\pi f_1 t) dt$$

$$N_2 = \int_0^T n(t) \cos(2\pi f_2 t) dt$$

$$r(t) = y(t) + n(t)$$

$$n(t) \sim \mathcal{N}(0, N_0)$$

$$E[N_1 N_2] = E \left[\int_0^T n(t) \cos(2\pi f_1 t) dt \cdot \int_0^T n(s) \cos(2\pi f_2 s) ds \right] = \int_0^T \int_0^T E[n(t) n(s)] \cos(2\pi f_1 t) \cos(2\pi f_2 s) dt ds$$

$$E[n(t) n(s)] = N_0 \delta(t - s)$$

$$E[N_1 N_2] = N_0 \int_0^T \cos(2\pi f_1 s) \cos(2\pi f_2 s) ds = \frac{N_0 T}{2} \text{sinc}(2(f_2 - f_1)T) \quad E[N_1^2] = E[N_2^2] = \frac{N_0 T}{2}$$

$$\text{Var}[N_1 - N_2] = E[N_1^2] + E[N_2^2] - 2E[N_1 N_2] = N_0 T (1 - \text{sinc}(2(f_2 - f_1)T))$$